FinTech Co – Technology Architecture Strategy

Contacts: Jeff Tijssen, Adam Davis, Ryan Garner, Marc Hochar, Reina Tabbara

May 2022

BAIN & COMPANY

All logos, trademarks, and brand names used belong to their respective owners
This information is confidential and was prepared by Bain & Company solely for the use of our client; it is not to be relied on by any 3rd party without Bain's prior written consent

Bain Approach

This deck is a part of a larger compendium. To have a look at more IPs from this case, search for the below case titles on Iris

POVs and Market Overviews

- Consumer Preferences towards Financial Services Needs POV (KSA, 2022)
- Consumer Preferences towards Financial Services Needs POV (UAE, 2022)
- SME Banking Needs POV (KSA, 2022)
- Digital Attacker Benchmarks and Financial Services Market Trends POV (KSA and UAE, 2021)
- Retail and SME Banking Overview (KSA, 2020)

Case examples

- FinTech Co – Roadmap to build a Digital Attacker
- **FinTech Co – Technology Architecture Strategy**
- Fintech Co – Regulatory Considerations and building BaaS Agreement with Bank Co
- Fintech Co – Building a Digital Attacker Operating Model

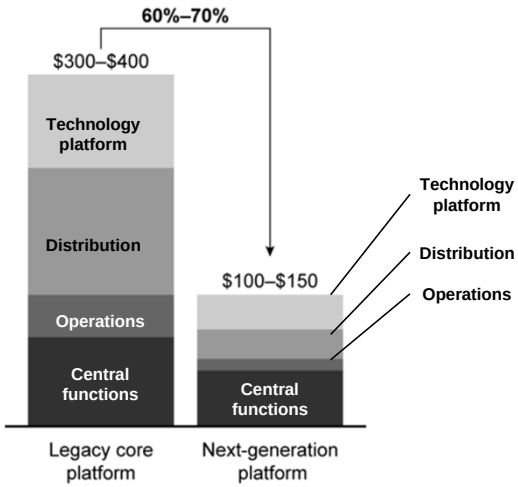
This information is confidential and was prepared by Bain & Company solely for the use of our client; it is not to be relied on by any 3rd party without Bain's prior written consent

Technical strategy have catalyzed the economics of digital attackers and neobanks globally

- Modern technology stacks have **changed the economics** relating to **set up and run costs** that it has catalyzed new Neobanks and Fintech's globally
- A digital attacker u**nderpinned by cutting edge technology** could have a cost base **60-70% lower than a mid-tier traditional** bank
- From a servicing perspective, attackers serve **2-3x as many customers per employee as legacy** core platform based technology stacks

Technology Platform	• **Lower unit cost** from microservices architecture and rationalized single core • **Software as a service** vendors take on **upgrades and API based open architectures** to **simplify the 'plug-in'** and **update processes**
Distribution	• Transition to **digital self-service** and **digital sales** function • **Branchless futures**
Operations	• **Know your customer** and **fraud** solutions that **reduce manual intervention**
Central functions	• Rationalized product portfolio • Automation tools • Clean **general ledger connections**, **single core**, **real-time** data for reconciliations

Annual cost per retail customer ($)



Source: Bain & Company benchmark data and experience, Thought Machine experience with contributions from Personetics, Onfido TruNarrative and Form3

This information is confidential and was prepared by Bain & Company solely for the use of our client; it is not to be relied on by any 3rd party without Bain's prior written consent

There are 6 potential architectural principles we need to follow to build the digital attacker

Microservice Architecture

- Design your services as microservice architectures - small modular services
  - Allows different parts of the system to be scaled in and out based on demand
  - Allows parts of the system to be upgraded in isolation

Event Driven

- Model your system as a stream of immutable events
  - Valuable for fault-finding, audit and scaling

Cloud Native

- Applications designed to take full advantage of container orchestration and modern cloud platforms, whilst remaining agnostic towards hosting vendors

Automated and Immutable

- Put in place full automation around the build and deployment, creating artefacts that are versioned and immutable through to Production
  - Allows you to perform Production changes and release new features rapidly, and with confidence.

API First

- All services should be built with well-defined APIs
  - All data changes should be made through these APIs, with standard controls to ensure both security and scaling

SaaS

- Don't look to build standard functions when they're readily-available, instead look to favour paying for support licences where necessary (rather than enterprise product licences which can tie you in)

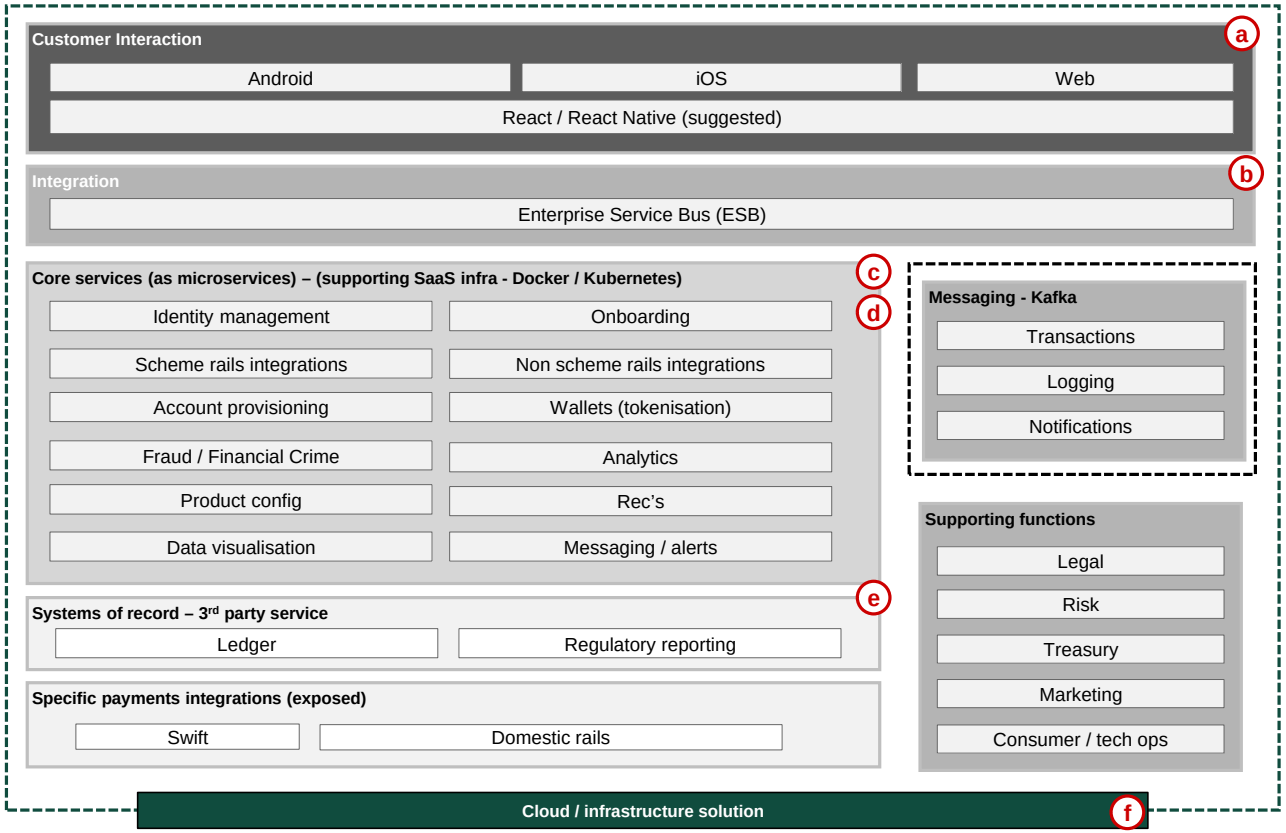
Design for KSA - The design of the technical stack should be based on the core market, taking into consideration the data rules and requirements of SAMA, and how we might replicate the architecture in other markets

Note: API – Application Programing Interface, SAMA - Saudi Arabian Monetary Authority, KSA – Kingdom of Saudi Arabia

This information is confidential and was prepared by Bain & Company solely for the use of our client; it is not to be relied on by any 3rd party without Bain's prior written consent

A typical best in-class digital attacker application architecture has 6 main components

/EXAMPLE



Comments application architecture

a **Front end to be built and owned by Fintech** to ensure flexibility, ownership, and speed of change

b **ESB** - Enterprise Service Bus implements a communication system between mutually interacting software applications in a service-oriented architecture

c Core services with a **'plug and play' capability** allowing for **integrations to 3rd parties** and best of breed providers

d **Microservices** are recommended to ensure **modularity of the stack**, **isolation of issues** and ability to **change providers** easily

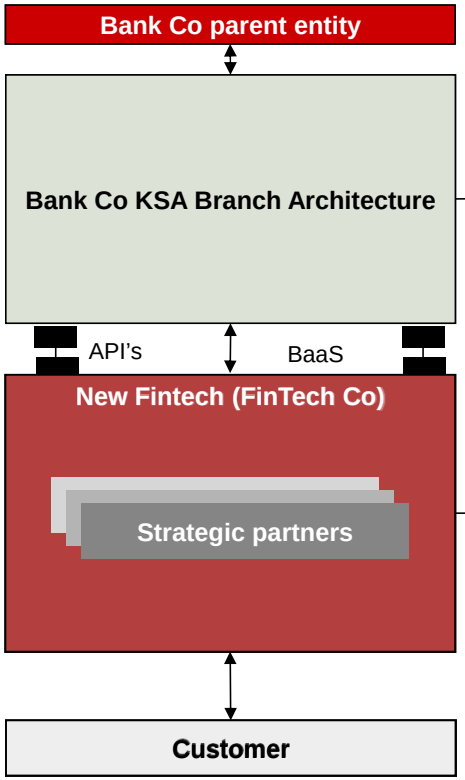
e **Core banking vendor** chosen will depend on the **final product roadmap**

f **Services** typically **hosted on the cloud** as it is more flexible and cost effective

This information is confidential and was prepared by Bain & Company solely for the use of our client; it is not to be relied on by any 3rd party without Bain's prior written consent

The choice of licensing option and its impact on technology

The license structure



The wider implications

- Responsible for provision of banking license
- Owner of balance sheet
- BaaS agreement with new Fintech
- Owns KYC / AML

- Separate SAMA Fintech license required
- Owner of customer data / UI & UX interfaces / Data and Analytics / brand and products / customer support and relationships
- New Fintech cannot be deposit taking directly or cannot underwrite loans

Key technology considerations

- Bank Co KSA Branch **architecture needs to be customized** to adopt the product suite of the Fintech
- **Back end will be owned by Bank Co KSA Branch** and thus will require enhancements:
  - **New core banking systems will be required** to fulfill product requirements (current core banking services corporate customers only)
  - **Regulated product enhancements that sit on Branch BS** (loans / deposits) **need to be built** and exposed as a service
  - Regulated **KYC, AML feeds will need to be built** to supply the fintech 'as a service'
  - **'Systems of record' layer will need** to be enabled to accept a RT feed from the fintech

- **New front end stack needs to be created to own customer interactions** and acquisition
  - API enabled to **integrate back into Bank Co KSA branch** architecture for reporting and monitoring purposes
  - **System of record layer should be replicated in the Fintech** to capture customer data for analytics and customer insights
  - **Separate reporting** line to be created to SAMA
  - **3rd party providers** who provide services **will have contracts with Fintech** (sometimes tripartite agreements will be needed)

This information is confidential and was prepared by Bain & Company solely for the use of our client; it is not to be relied on by any 3rd party without Bain's prior written consent

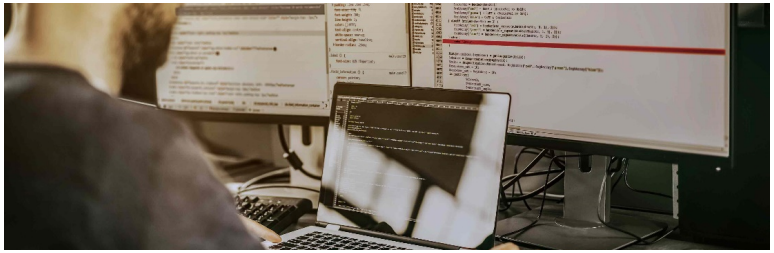
Bank Co KSA: For the Bank Co KSA stack, there are three potential options

	1 Use existing TCS with enhancement	2 Overlay back-end on top of existing TCS	3 Build new core for entire Bank Co KSA
Overview	• TCS is currently used in Bank Co KSA • **Re-use elements of TCS** where applicable and **buy missing functionalities** required for the digital attacker from TCS	• Re-use elements of **TCS where applicable** • **Buy missing functionalities** that are required for the digital attacker **from a different vendor** e.g. Mambu and connect to existing stack	• Build **new core banking system for entire Bank Co KSA** and migrate the legacy corporate bank to the new system as well
Pros	+ **Quick** time to market + Fit for purpose **if product suite is limited** and **roadmap is traditional**	+ **Fit for purpose** in supporting innovative CVPs + Allows Bank Co KSA to have a **separate system for** the **Fintech** and **do BaaS with other entities in the future** + **No impact** on **existing activities / customers** + **Higher agility** (with new vendor specialized in digital banks) + Allows for **Opex costs (SaaS) vs. Capex for TCS configuration** + **Maintenance** to be provided by the SaaS vendor	+ **Fit for purpose** in supporting innovative CVPs + Allows Bank Co KSA to **do BaaS with other entities in the future** + **Higher agility** (with new vendor specialized in digital banks) + Allows for **Opex costs (SaaS) vs. Capex for TCS configuration**
Cons	- **Requires Capex** for **configuration** and to **sustain volumes for retail banking** - **Basic product configurations** & limited cutting-edge digital technologies (e.g.: BNPL) - **Lacks API** enabled back-end infrastructure (modern front-end will be difficult to integrate) - Requires **in-house maintenance**	- **Integration can get complex** and **require time**	- **Additional time to market required** - **Migration is complex** and will **interrupt current business** - Requires **significant capital**

Final decision to be made once detailed architecture design is completed

This information is confidential and was prepared by Bain & Company solely for the use of our client; it is not to be relied on by any 3rd party without Bain's prior written consent

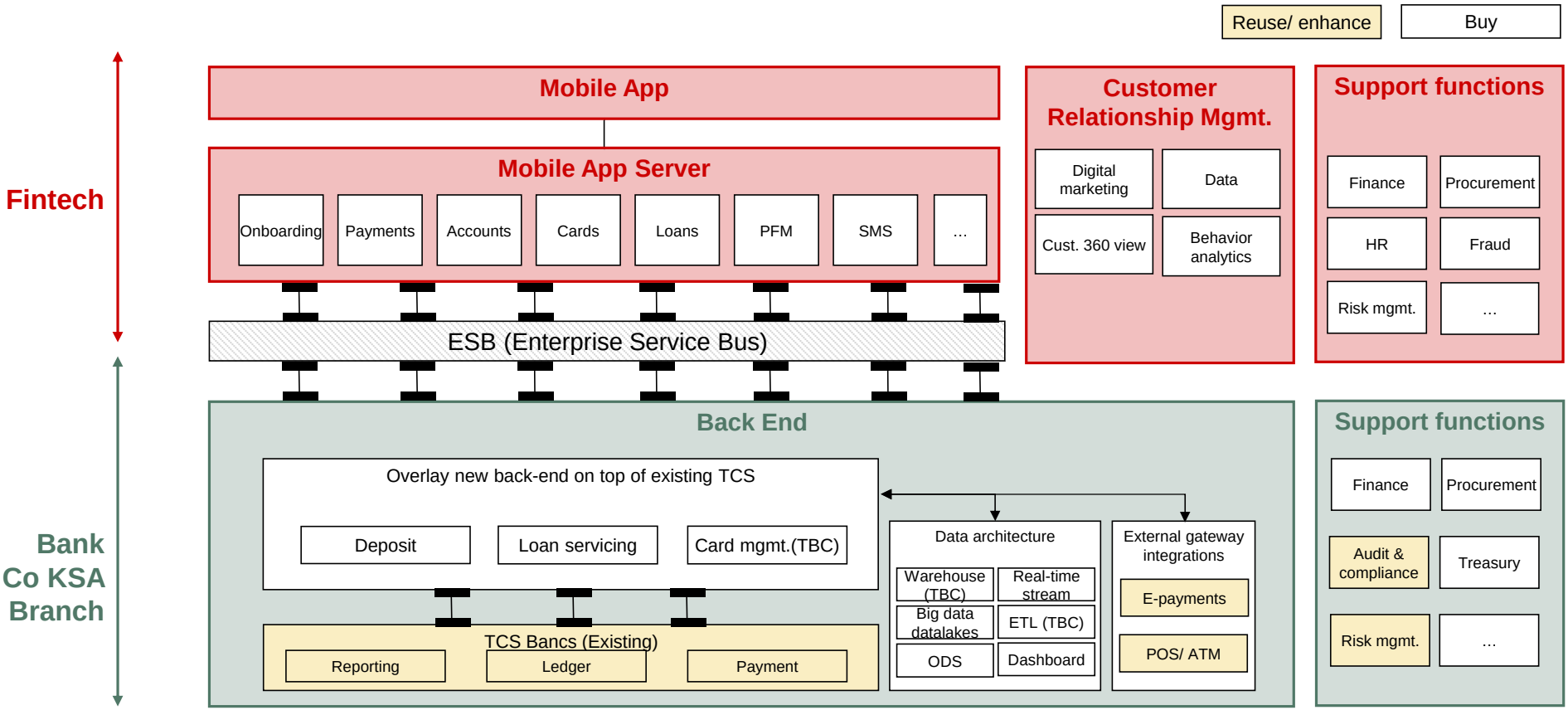
New Fintech: To implement the front-end (mobile app) there are two options either to buy or to build

	Buy and customize	Build
		
Pros	+ **Faster** time to market + **Easier to maintain** due to vendor maintenance support + **Solutions offered have been tried and tested**	+ Added **differentiation and flexibility** as the front-end is built from scratch + **Team capabilities can grow by working closely with vendors,** thereby retaining knowledge through build and run
Cons	- **Significant upfront capital** spend - Might require **multiple customizations** to fit the features that are required for FinTech Co **Recommended Option**	- Will take **significant time to build** from scratch - **Limited** development **talent** in the market and **in-house** (not an internal core competency) - **Difficult to maintain** as maintenance activities will need to be done by in-house resources

This information is confidential and was prepared by Bain & Company solely for the use of our client; it is not to be relied on by any 3rd party without Bain's prior written consent

Potential Architecture Diagram for FinTech Co

/ TO BE REFINED DURING BUILD PHASE / DIRECTIONAL



This information is confidential and was prepared by Bain & Company solely for the use of our client; it is not to be relied on by any 3rd party without Bain's prior written consent

The product capabilities can be translated into 12 technical capabilities with limited level of re-use from Existing Bank Co KSA Branch

Reuse and enhance | Non Available | Cannot be re-used
Bank Co KSA Branch | Fintech | TBC

/ TO BE REFINED DURING BUILD PHASE / DIRECTIONAL

1. Innovative customer-facing channels

- Digital (mobile, web, tablet, etc.)
- **Remote (call center, self service, etc.)**
- **Ecosystem partners access (Open API, monetization API)**

2. Customer relationship management

- **Digital Marketing**
- **Campaign mgmt.**
- **Customer 360 view**
- **Forward looking insights**
- **Customer feedbacks**
- **Real-time marketing**
- **Customer data & loyalty**
- **User behavior analytics**

3. Access management gateway

- **Customer IDM**
- API security
- 2FA/ OTP capability
- Biometrics / E-KYC

4. Customer centric platform

- **Omni channel business rule engine**
- Document management
- **Enterprise workflow engine**
- **Digital Signature**

5. API gateway

Micro-services

6. Highly automated product processing

Central product processing
Business rule engine

Product processing

- Deposit
- Payment
- Card mgmt.
- Loan servicing/admin.
- **Loyalty**
- **PFM**

- **E-statement solution**

7. External gateways access

- POS/ ATM
- E-payments
- Payment network
- **Card scheme**
- SMS/ Push notification
- Social media

External parties

- …
- …
- …
- …
- …
- …
- …
- …

External data

12. Modern integration layer

8. Modern data architecture

- **Real-time stream analytics**
- **Big data datalakes**
- Reporting
- SAMA reporting
- **ODS**
- Data warehouse
- ETL
- **Dashboard**

9. Robust support functions

- **Legal**
- Audit & Compliance
- **Finance Treasury**
- **Proc-urement**
- Risk mgmt
- HR
- Enterprise Scheduler
- **Finance**

10. Security

- API secuity
- SIEM
- **FIM**
- PAM
- Security Testing / Assessment

11. Infrastructure

- Data center
- Network
- Other

Note: ODS - operational data store
Source: Bain analysis

This information is confidential and was prepared by Bain & Company solely for the use of our client; it is not to be relied on by any 3rd party without Bain's prior written consent

At the beginning of the build phase, we will need to run vendor selection across the stack

EXAMPLE PROVIDERS

Existing Bank Co vendors

Front-end	BACKBASE, React Native
Digital onboarding / KYC providers	aion, ShuftiPro, hooyu, JUMIO, Kastle, IDEMIA, onfido, Daon
PFM	STRANDS, meniga, MX, CSI, Geezeo
Loyalty	icc fintech, AIMIA LOYALTY SOLUTIONS, bond BRAND LOYALTY, COLLINSON, EPSILON
Real time marketing	evergage A Salesforce Company, BOXEVER, epi, W.UP
AI Chatbots	ACTIVE.Ai, Finn AI, Kasisto
Product Config	ZAFIN
CRM Providers	salesforce, Microsoft Dynamics
Core Banking	MAMBU, TCS BaNCS, TEMENOS, ORACLE
Cloud providers	موبايلي mobily, STC, DETASAD, ORACLE

Development vendors

- FINANTEQ — Front-end developers
- software AG — API developers
- vacuumlabs — E2E development capability
- endava — E2E development capability

This information is confidential and was prepared by Bain & Company solely for the use of our client; it is not to be relied on by any 3rd party without Bain's prior written consent